

Problem Statement — NYC Airbnb Market Analysis

Does regulatory pressure from Local Law 18 leave a visible fingerprint in Airbnb's own listing data?

BACKGROUND

New York City's short-term rental market has undergone significant regulatory change in recent years. **Local Law 18** (the Short-Term Rental Registration Law), enforced since September 2023, requires hosts offering stays under 30 nights to register with the city's Office of Special Enforcement (OSE). Hosts who don't register — or who avoid the requirement entirely by setting a 30-night-or-longer minimum stay — operate outside this regulatory framework.

This creates an interesting analytical question: **does this regulatory pressure leave a visible fingerprint in Airbnb's own listing data?** A dataset of 20,770 NYC Airbnb listings, scraped with 22 attributes per listing (price, location, room type, license status, availability, review activity, and more), offers a way to investigate this directly — alongside the more conventional questions a market dataset like this naturally invites.

PROBLEM STATEMENT

Beyond understanding general pricing and market structure, this project investigates whether **regulatory avoidance behavior is statistically visible** in listing-level data, using minimum night requirements and license status as the primary lens — while also addressing the practical, real-world data quality problems (corrupted identifiers, missing values, inconsistent formatting) that stand between a raw scraped dataset and reliable analysis.

OBJECTIVES

1. **Clean and validate** a real-world, imperfect dataset — including corrupted numeric identifiers, missing values with non-random patterns, duplicate records, and text-corrupted numeric fields.
2. **Characterize the NYC Airbnb price distribution** — including its skew, spread, and outliers — and identify what drives price variation across boroughs and room types.
3. **Test whether license status and minimum-stay requirements are related**, as a proxy for whether hosts are structuring listings to avoid short-term rental registration.
4. **Identify listing activity patterns** — including a suspected "ghost listing" phenomenon visible in the `availability_365` distribution — that may reflect inactive or long-term-rental listings still present on a short-term rental platform.
5. **Deliver findings through three complementary tools**: a documented Python/pandas cleaning and EDA pipeline, a set of SQL queries for relational analysis, and an interactive Power BI dashboard for non-technical stakeholders.

KEY QUESTIONS

- How is price distributed across NYC's boroughs and room types, and how much does it skew from a handful of extreme listings?
- What share of listings default to a 30-night minimum stay, and does this correlate with a listing's license status?
- What proportion of listings hold a formal short-term rental license, versus operating unlicensed or under a legal exemption?
- Do listings show signs of being inactive ("ghost listings") based on availability and recent review activity?
- What data quality issues exist in the raw dataset, and how do they affect the reliability of any conclusions drawn from it?

SCOPE

In scope: exploratory and descriptive analysis of listing-level attributes (price, location, room type, license status, minimum nights, availability, reviews) as captured in the dataset's snapshot in time.

Out of scope: causal claims about *why* individual hosts choose specific licensing or minimum-stay strategies (the dataset shows correlation, not host intent); time-series analysis of how the market changed before/after Local Law 18 enforcement, since this dataset represents a single point in time rather than a longitudinal sample; and legal or compliance advice — all regulatory framing is descriptive context, not a legal assessment of any individual listing.

TOOLS & METHODOLOGY

STAGE	TOOL	PURPOSE
Data cleaning & EDA	Python (pandas, matplotlib, seaborn)	Inspect, clean, and explore the raw dataset; univariate and bivariate analysis
Relational querying	PostgreSQL	Store the cleaned dataset; practice and demonstrate SQL analysis independent of Python
Dashboard & visualization	Power BI	Build an interactive, stakeholder-facing view of key findings

DATASET

- **Source:** NYC Airbnb listings dataset (`datasets_newyork.csv`)
- **Size:** 20,770 rows × 22 columns
- **Key fields:** `price` , `neighbourhood_group` , `room_type` , `minimum_nights` , `license` , `availability_365` , `number_of_reviews` , `reviews_per_month` , `rating` , `bedrooms` , `beds` , `baths`